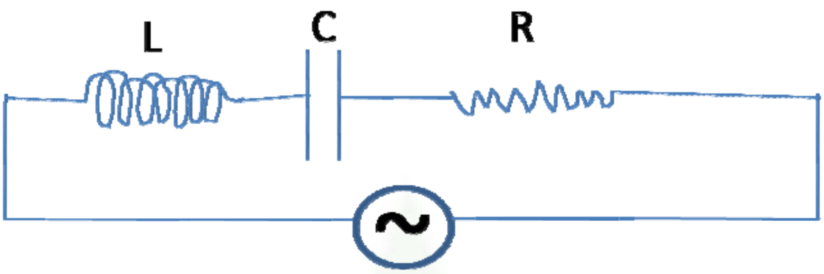
L-C-R circuit:



The impedance of the circuit

$$z = \sqrt{R^2 + (X_L - X_C)^2} = \sqrt{R^2 + \left(\omega L - \frac{1}{\omega c}\right)^2}$$

The emf may lead or lag the current depends on the values of ω,L & C

(1) If $\omega L > \dfrac{1}{\omega c}$ then emf leads the current by phase angle ϕ

where $\tan\phi = \dfrac{\left(\omega L - \frac{1}{\omega c}\right)}{R}$

(2) If $\omega L < \dfrac{1}{\omega c}$ the emf lags the current by a phase angle ϕ

where $\tan\phi = \dfrac{\left(\frac{1}{\omega c} - \omega L\right)}{R}$